META-LABELING

Meta-Labeling

The primary model sets the side; the secondary model decides whether to trade and how much

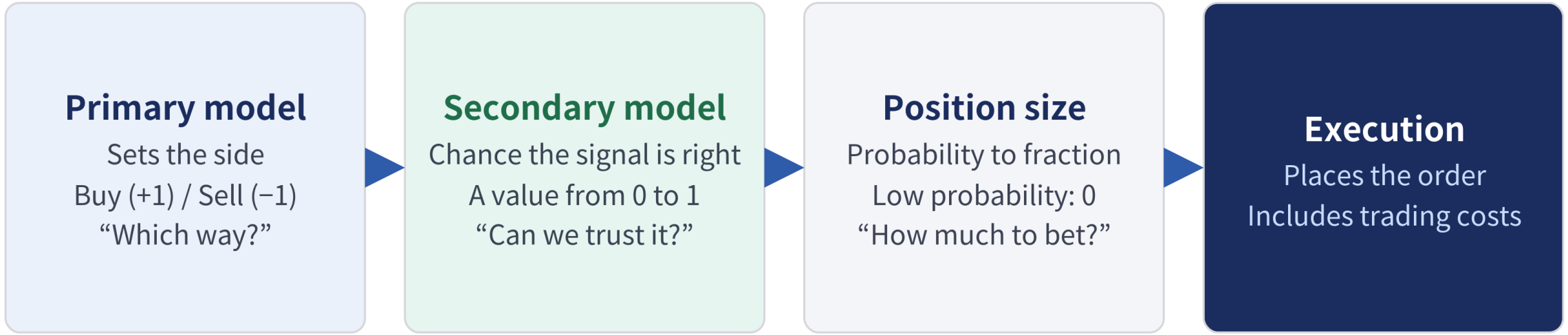
Meta-Labeling — Triple Barrier, Precision·Recall, Calibration and Bet Sizing

One example — a primary model with only 40 winners in 100 signals — followed from start to finish,
covering triple-barrier labels, meta-labels, probability calibration, position sizing and recent research.

Meta-labeling splits the side decision and the size decision across two models

The primary model picks which side to bet on; the secondary model decides whether to bet and how much (López de Prado, 2018)

The primary model sets the side; the secondary model decides whether to trade and the size



The primary model can be anything — moving-average crossover, arbitrage rule, fundamental quant, another ML model

The primary model casts a wide net; the secondary model picks out the real signals

Think of airport security — the inspector opens only the bags the sniffer dog barked at

Primary model = sniffer dog

Barks at anything suspicious
Misses almost no bags (recall ↑)
Many false barks too (precision ↓)



Secondary model = inspector

Opens only bags the dog barked at
Tells real finds from false alarms
Filters false barks to raise precision

The inspector never sees bags the dog ignored — the secondary model creates no new signals

01

PART 1 · Foundations

Labels and the Triple Barrier

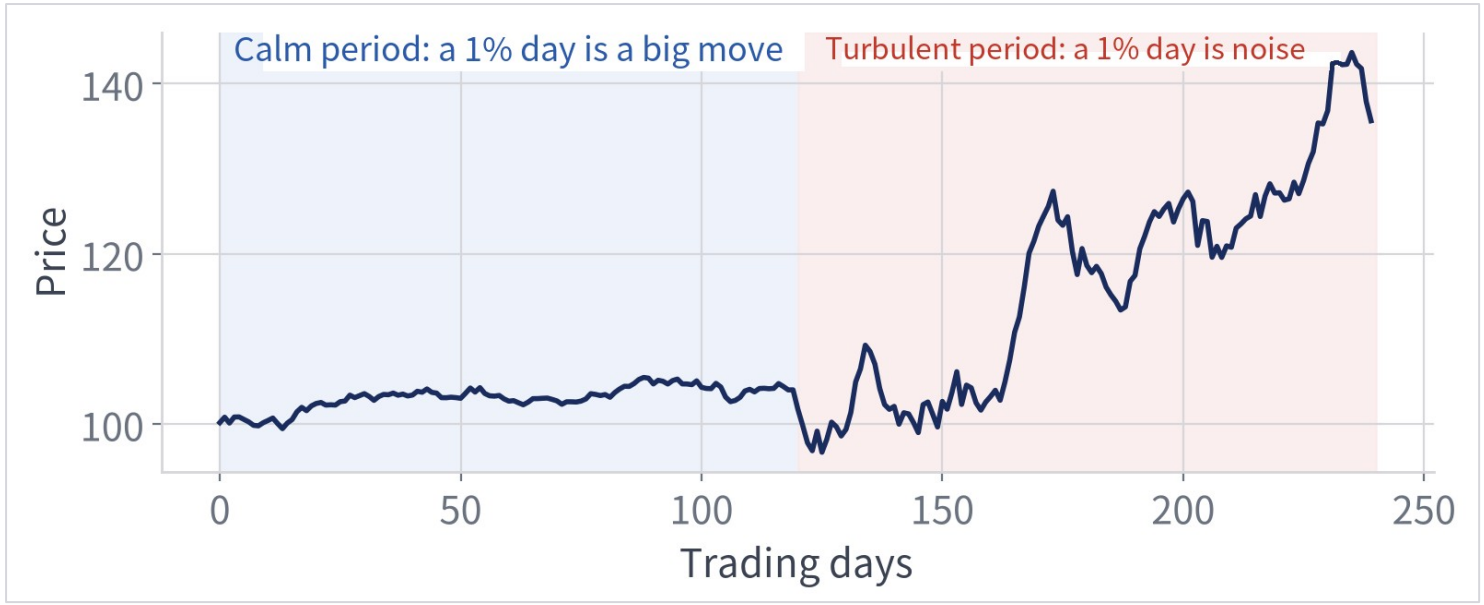
How to build the answers for machine learning from financial data

Fixed-horizon labels ignore volatility that changes over time

Label = the answer given to a supervised model. In finance the answer is ‘was this trade profitable?’

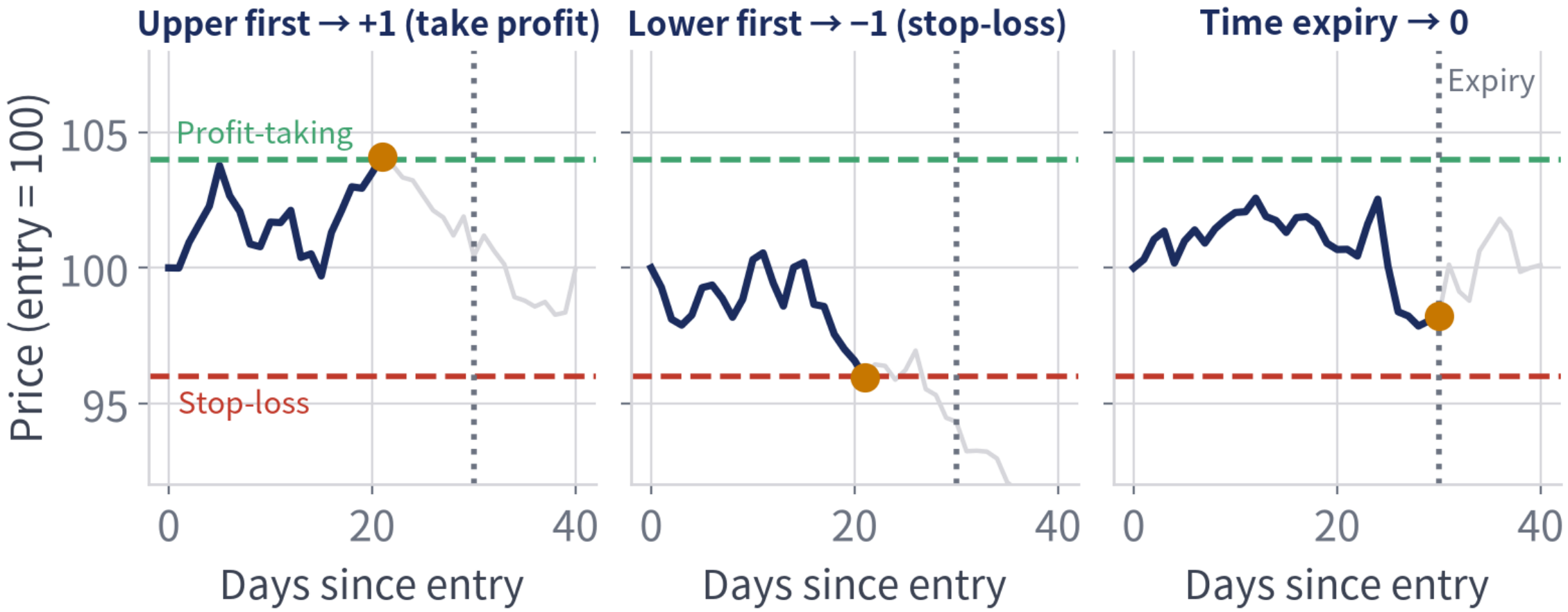
The problem with the simplest method

- It is tempting to fix the horizon and threshold, e.g. ‘1 if the 10-day return exceeds +1%’
- A 1% move means something entirely different in a calm period and in a turbulent one
- A trade that breaches the stop-loss mid-horizon and then recovers is still recorded as a win



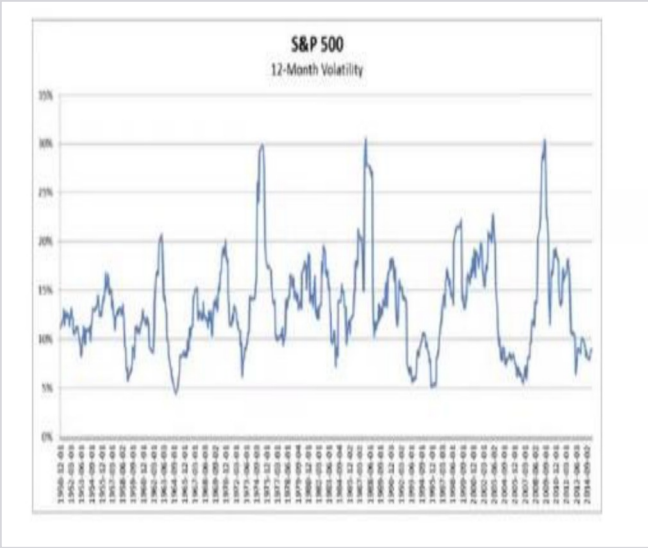
The triple barrier sets the outcome by whichever of three lines is touched first

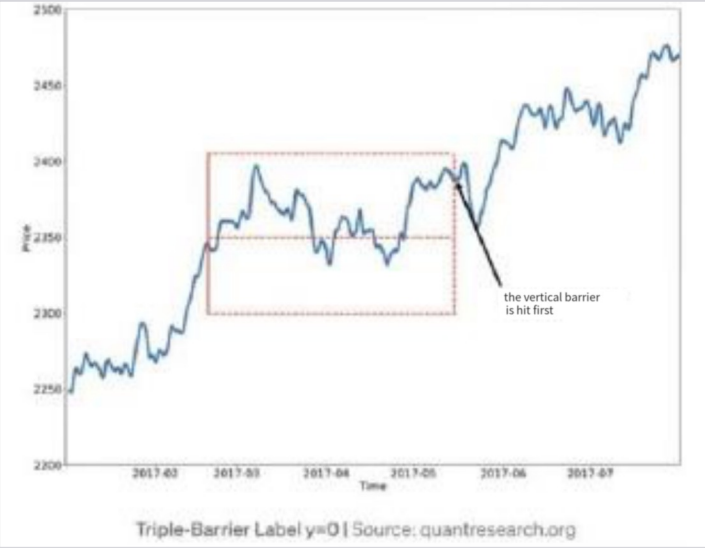
Top: profit-taking barrier (horizontal), bottom: stop-loss barrier (horizontal), right: holding limit (vertical) — triple barrier



Barrier widths are set to the volatility of the moment

Left: S&P 500 12-month volatility (5–30%), right: a barrier box drawn on actual prices (quantresearch.org)





Why scale to volatility

- Volatility varies by up to six times across periods. Setting the width as ‘a multiple of daily volatility’ makes the target equally hard in any period
- The red box in the right figure is the three barriers of one trade. The outcome is set where the price first leaves the box

02

PART 2 · Principles

The Principle of Meta-Labeling

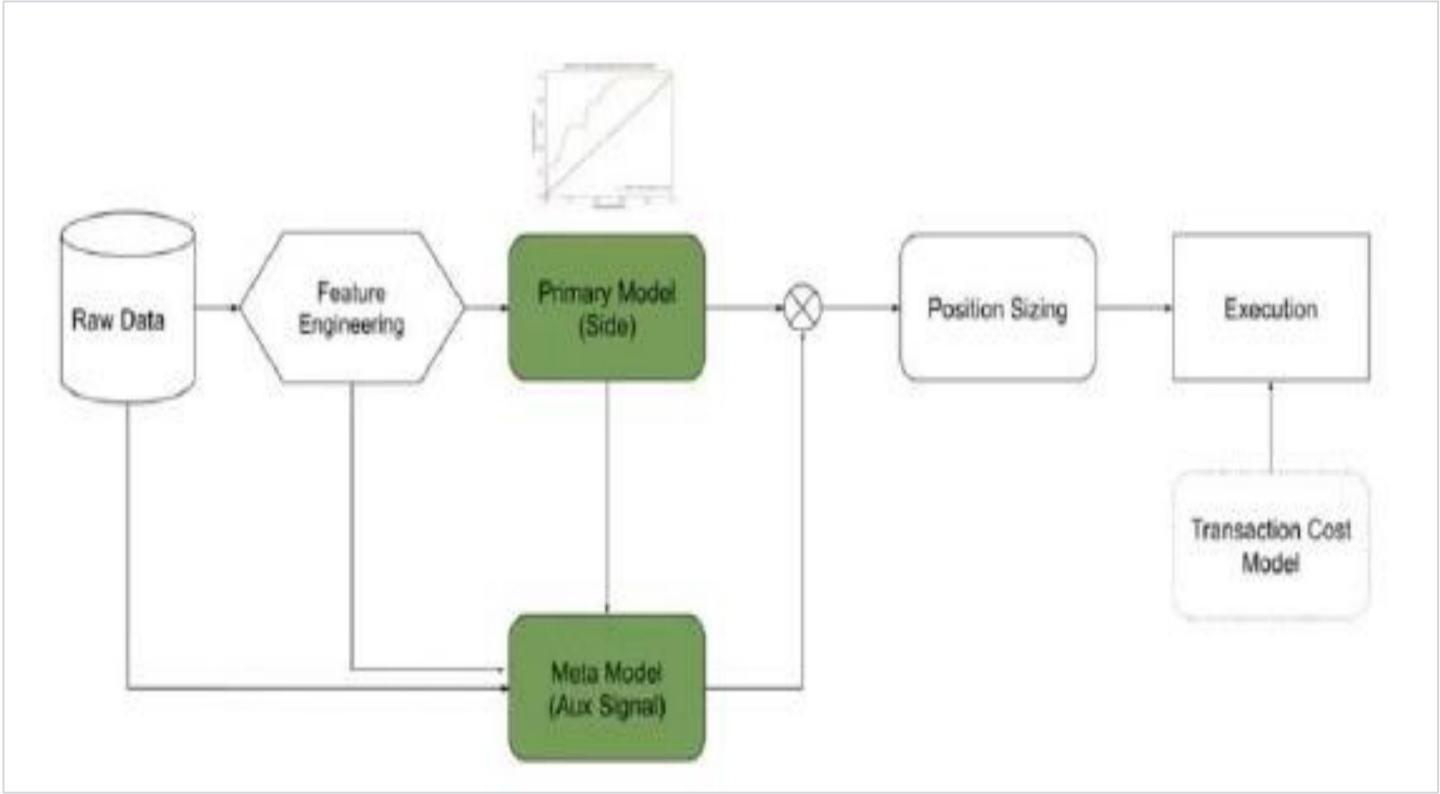
The second model learns only ‘will this signal be right?’

The secondary model predicts whether the primary signal was right

To trade or not to trade — the ⊗ in the circle means the primary side is multiplied by the secondary decision

Division of roles

- Primary model (side): outputs either buy +1 or sell −1
- Secondary model (meta, aux signal): outputs whether that signal will be right (1) or wrong (0)
- The primary model can be arbitrage, technical analysis, traditional quant or an ML model



The meta-label is 1 if trading in the primary direction made a profit

The same ‘upper barrier first’ is a profit (1) on a buy signal and a loss (0) on a sell signal — side and outcome are read together

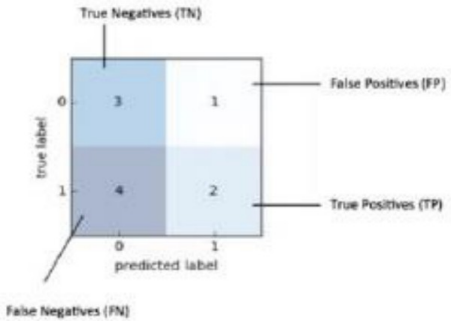
Signal	Primary side	Barrier outcome	Trade P&L	Meta-label
①	Buy +1	Upper first (price ↑)	Profit	1
②	Buy +1	Lower first (price ↓)	Loss	0
③	Sell −1	Lower first (price ↓)	Profit	1
④	Sell −1	Upper first (price ↑)	Loss	0
⑤	Buy +1	Time expiry (nearly flat)	P&L ≤ 0	0

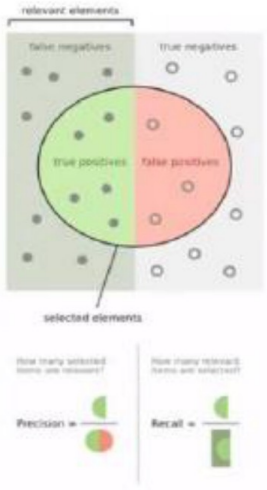
Precision measures trade quality; recall measures how few opportunities are missed

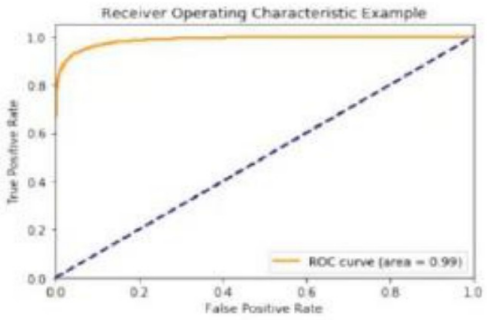
TP: traded and profitable, FP: traded at a loss, FN: not traded, but the signal would have been profitable

$$\text{Precision} = \frac{\text{profitable signals among those traded}}{\text{signals traded}} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{\text{profitable signals that were traded}}{\text{profitable signals}} = \frac{TP}{TP + FN}$$

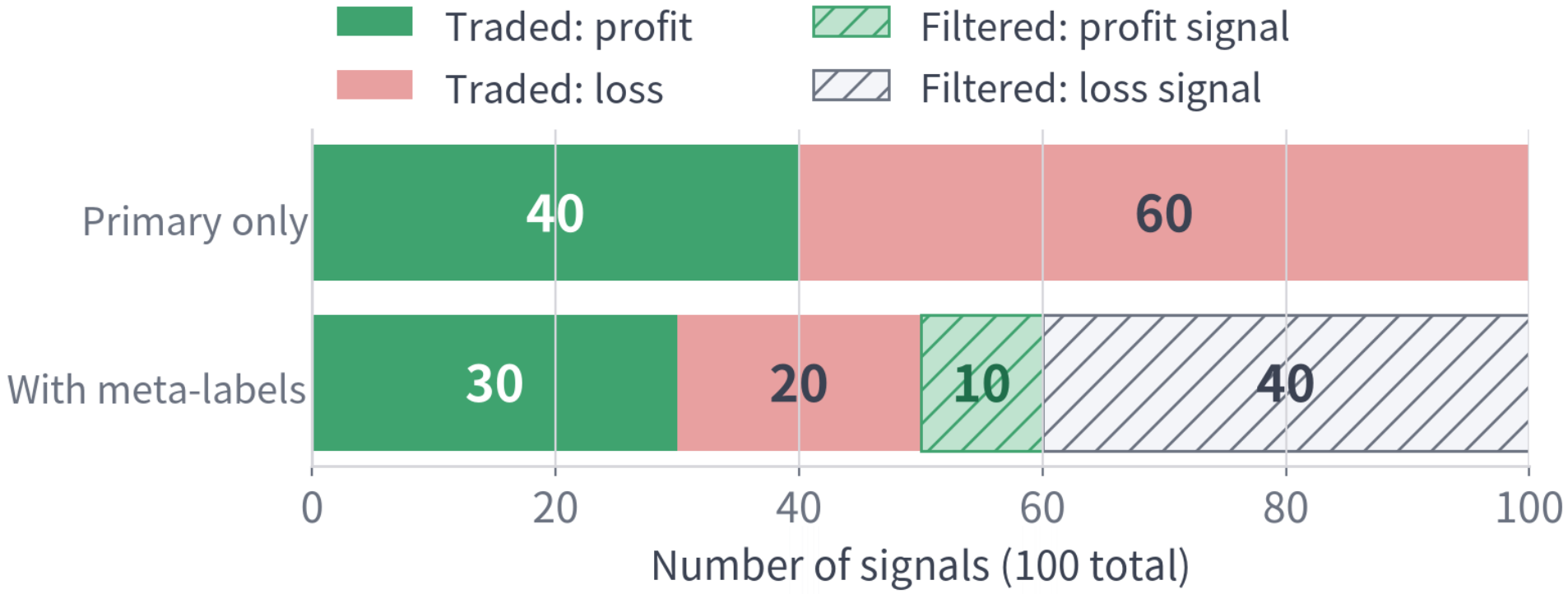






Meta-labeling gives up a little recall to gain a lot of precision

Example: 40 of the primary model's 100 signals are profitable — the secondary model keeps 50 and filters out 50 (※ illustrative)



Of the 50 kept, 30 are profitable (precision 60%); 30 of the 40 profitable signals are retained (recall 75%)

The primary model handles recall; the secondary model handles precision

How meta-labels work — one model does not try to achieve both

Primary model: high recall

- Low precision is acceptable. The threshold is lowered so that few profit opportunities are missed
- In the example, 100 signals of which 40 are profitable — precision 40%

Secondary model: better precision

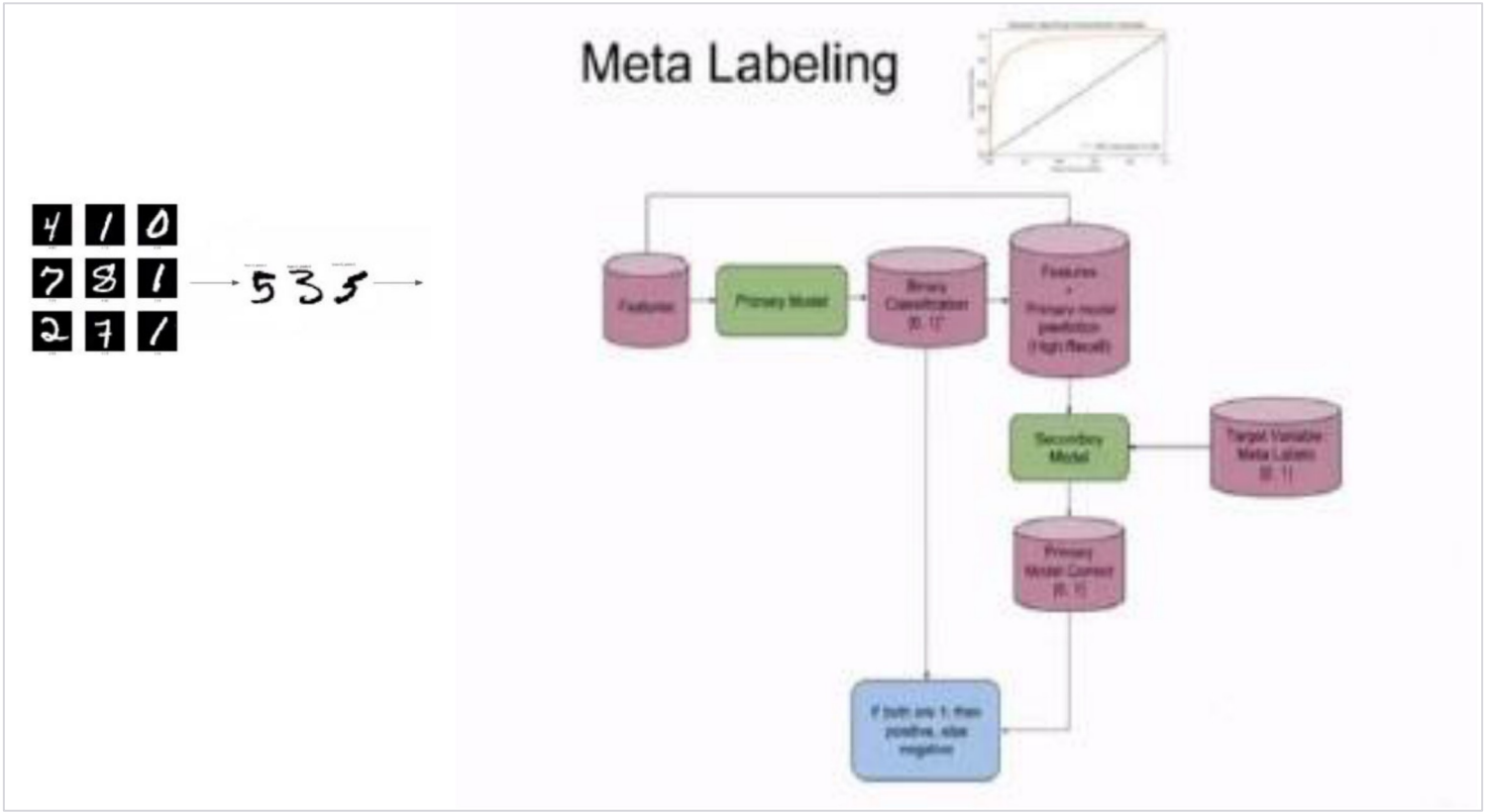
- Meta-labels are learned only on signals where the primary model said ‘trade’ (Positive)
- Filtering losing signals lifts precision to 60%, at the cost of missing 10 profitable signals

Learn the structure first on handwritten digit classification

A structural example of meta-labels — the secondary model learns ‘was the primary model right?’

Flow

- The primary model performs binary classification (0/1) from the features
- Features + primary prediction are the secondary inputs; ‘was the primary right (0/1)?’ is the label
- Final verdict: positive if both are 1, otherwise negative



Meta-labels cut false positives from 192 to 35

Confusion matrix [[TN, FP],[FN, TP]] — left: primary model, right: with meta-labels (1,902 validation samples)



Base Model Metrics:

	precision	recall	f1-score	support
False	0.95	0.94	0.94	892
True	0.94	0.96	0.95	1010
micro avg	0.95	0.95	0.95	1902
macro avg	0.95	0.95	0.95	1902
weighted avg	0.95	0.95	0.95	1902

Confusion Matrix
[[700 192]
[11 999]]
Accuracy: 0.8933

Meta Label Metrics:

	precision	recall	f1-score	support
False	0.95	0.96	0.95	892
True	0.96	0.95	0.96	1010
micro avg	0.96	0.96	0.96	1902
macro avg	0.96	0.96	0.96	1902
weighted avg	0.96	0.96	0.96	1902

Confusion Matrix
[[857 35]
[47 963]]
Accuracy: 0.9569

※ The table below uses the confusion matrix. The Base Model report in the capture (precision and recall 0.94–0.96) has the same sample counts (892/1,010) but does not match the matrix (accuracy 0.8933), so it appears to come from different predictions.

Metric (from the confusion matrix)	Primary model	With meta-labels
False positives FP (false alarms)	192	35
False negatives FN (misses)	11	47
Precision TP/(TP+FP)	999/1,191 = 83.9%	963/998 = 96.5%
Recall TP/(TP+FN)	999/1,010 = 98.9%	963/1,010 = 95.3%
Accuracy	89.3%	95.7%

Even a small gain in precision greatly improves risk metrics

Bollinger Band primary model + meta model, out-of-sample period 2018-01-04 to 2019-01-28 (※ table values as in the source capture)

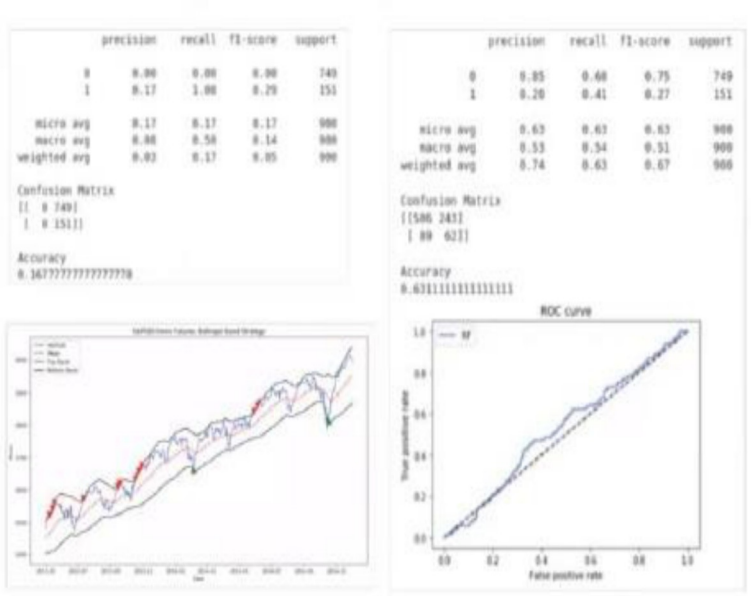


Table 1: Out-of-sample (2018-01-04 : 2019-01-28)

	Primary Model	Meta Model
Annual return		
Cumulative returns	19.7%	39.6%
Annual volatility	95.0%	56.7%
Sharpe ratio	0.65	0.82
Calmar ratio	0.29	0.96
Max drawdown	-61.9%	-36.8%
Daily value at risk	-11.7%	-7.0%

Out-of-sample performance	Primary	Meta
Cumulative return	19.7%	39.6%
Annual volatility	95.0%	56.7%
Sharpe ratio	0.65	0.82
Calmar ratio	0.29	0.96
Maximum drawdown	−61.9%	−36.8%
Daily VaR	−11.7%	−7.0%

Trade precision rose only from 17% to 20%, but filtering losing trades sharply reduced drawdown and volatility

03

PART 3 · Position Size

From Probability to Position Size

The secondary model's probability tells us not only whether to trade but how much to bet

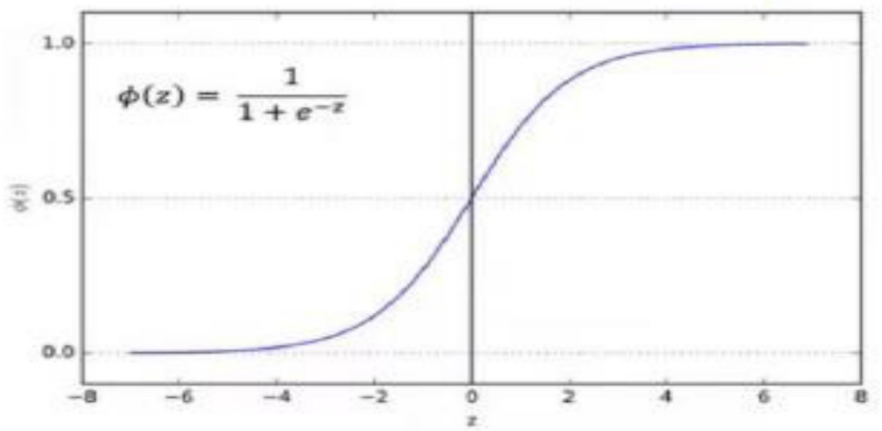
The secondary model's output is useful only as a probability between 0 and 1

The logistic (sigmoid) function φ(z) turns a model score z into a probability

$$\phi(z) = \frac{1}{1+e^{-z}}$$

A well-calibrated classifier

- The larger the score z, the closer the probability is to 1; the smaller, the closer to 0
- Samples with predict_proba of 0.8 should actually be about 80% positive
- Only then can ‘probability 0.8’ be read as ‘80% chance of winning’ and used to set the bet size



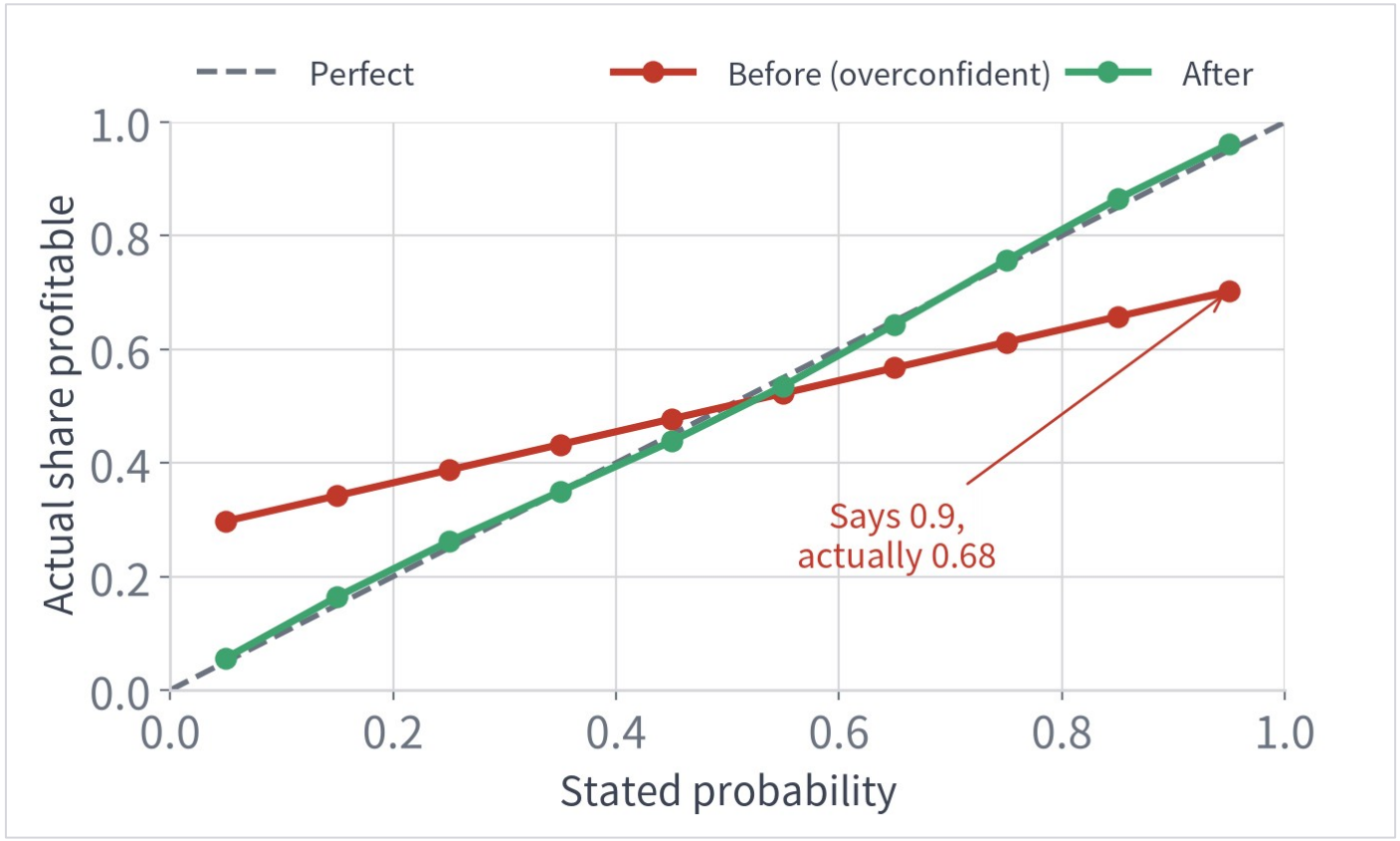
Example: a well calibrated (binary) classifier should classify the samples such that a predict_proba value of 0.8 should be approximately a 80% frequency of the positive class.

A model's stated probability easily departs from the actual frequency, so calibration is needed

Probability calibration curve — the closer to the diagonal, the closer stated probability and actual frequency (※ illustrative data)

Two calibration methods

- Platt scaling: fits a single sigmoid on top. Suits curves bent into an S shape
- Isotonic regression: fits a stepwise monotonic function. Good with ample data but can overfit
- Calibrate on data not used for training



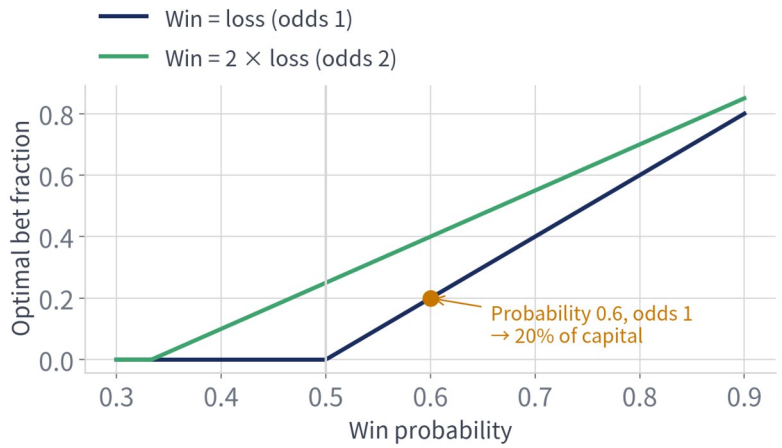
The Kelly formula gives the optimal bet fraction from win probability and odds

Kelly criterion — the bet fraction that maximizes long-run wealth growth

$$\underbrace{f^*}_{\text{optimal bet fraction}} = \frac{\beta p - q}{\beta} \qquad \beta\text{: odds (win amount/loss amount)}, \quad p\text{: win probability}, \quad q = 1 - p$$

$$\beta = 1 \ (\text{win} = \text{loss}) \ \Rightarrow \ f^* = p - q = 2p - 1$$

$$\text{Investment setting:} \quad f^* = \frac{p}{\text{loss\%}} - \frac{q}{\text{profit\%}}$$



Link to meta-labels

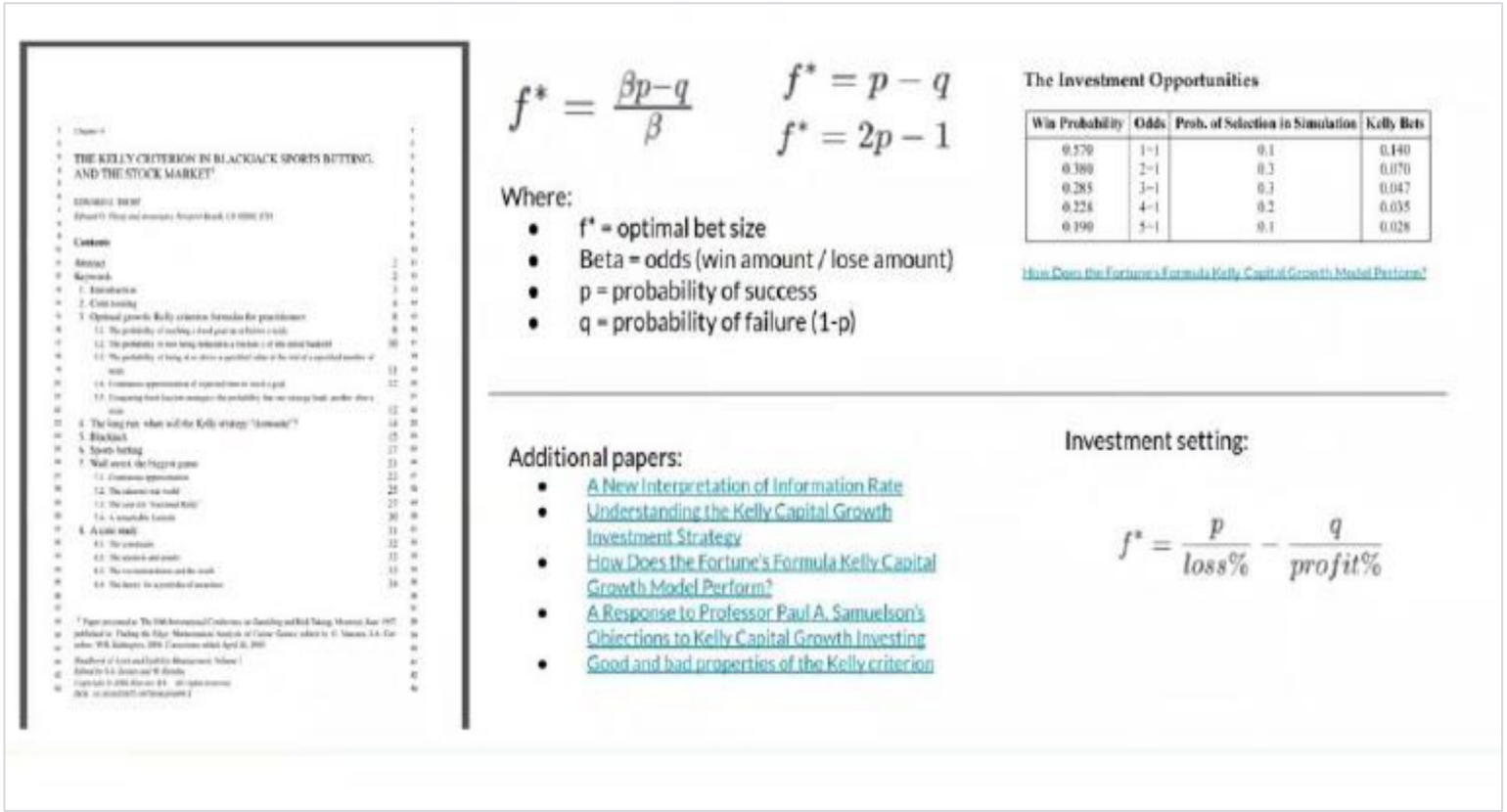
- Put the secondary model's probability in place of p. At odds 1, do not trade at 0.5 or below
- At probability 0.6 and odds 1, bet 20% of capital

The Kelly formula crossed from blackjack to the stock market

Thorp's paper (The Kelly Criterion in Blackjack, Sports Betting, and the Stock Market)

Reading the table

- Win probability 0.570 at 1:1 odds gives a Kelly fraction of 0.140 (2×0.570 − 1)
- The higher the odds, the lower the win probability at which a bet is worthwhile
- In practice, estimation error means betting half of Kelly or less



THE KELLY CRITERION IN BLACKJACK SPORTS BETTING, AND THE STOCK MARKET*

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* Paper presented at The 10th International Conference on Gambling and Risk Taking, Montreal, June 1987; published in: Finding the Edge: Mathematical Analysis of Casino Games, edited by O. Vancura, J.A. Cot-trell, W.R. Eadington, 2000. Conventions added April 26, 2005.

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$$f^* = \frac{\beta p - q}{\beta} \qquad f^* = p - q \qquad f^* = 2p - 1$$

Where:

- f* = optimal bet size
- Beta = odds (win amount / lose amount)
- p = probability of success
- q = probability of failure (1-p)

The Investment Opportunities

Win Probability	Odds	Prob. of Selection in Simulation	Kelly Bets
0.570	1-1	0.1	0.140
0.380	2-1	0.2	0.070
0.285	3-1	0.3	0.047
0.228	4-1	0.2	0.035
0.190	5-1	0.1	0.028

How Does the Fortune's Formula Kelly Capital Growth Model Perform?

Additional papers:

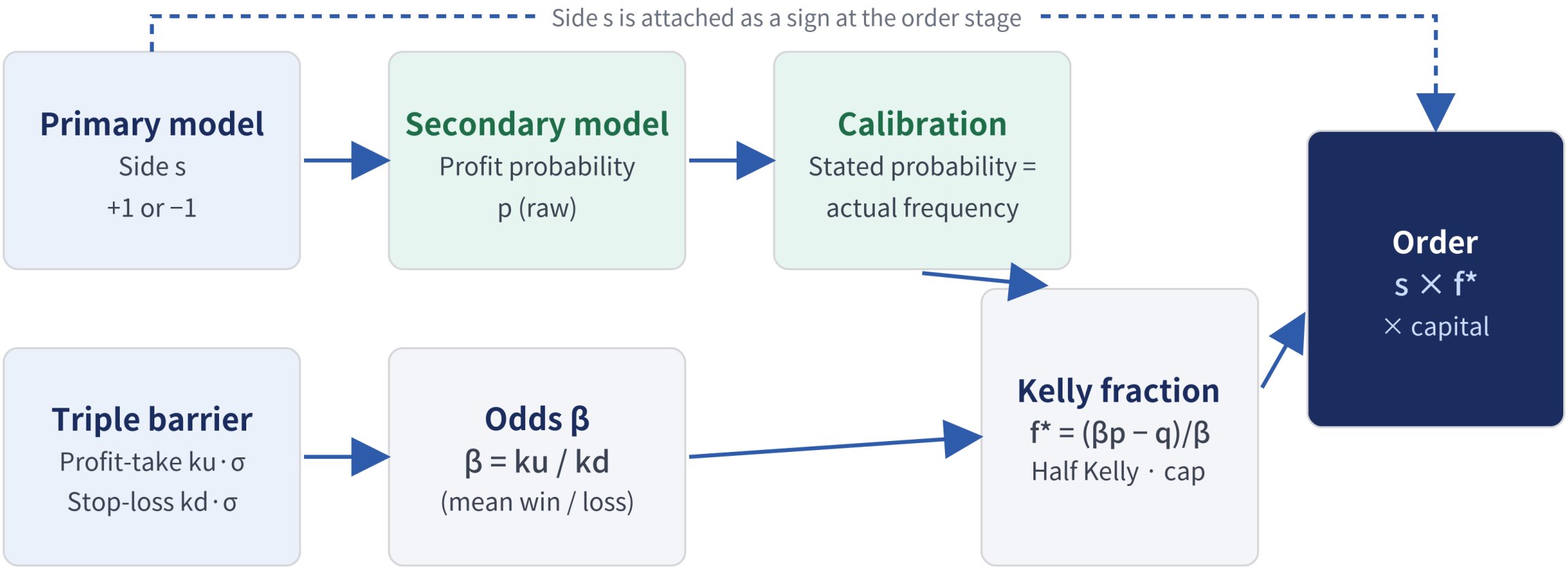
- A New Interpretation of Information Rate
- Understanding the Kelly Capital Growth Investment Strategy
- How Does the Fortune's Formula Kelly Capital Growth Model Perform?
- A Response to Professor Paul A. Samuelson's Objections to Kelly Capital Growth Investing
- Good and bad properties of the Kelly criterion

Investment setting:

$$f^* = \frac{p}{loss\%} - \frac{q}{profit\%}$$

The secondary model's probability fills the win-probability slot in the Kelly formula

The primary model gives the sign, the triple barrier the odds, the secondary model the win probability — together, the order size



The Kelly fraction is set by probability and odds; the side is attached as a sign

Maps each symbol of the Kelly formula to the stage of the meta-labeling pipeline it comes from

$$f^* = \frac{\beta \overbrace{p}^{\text{secondary probability}} - \overbrace{(1-p)}^{q}}{\underbrace{\beta}_{\text{odds}}}$$

$$\text{Order} = \underbrace{s}_{\text{primary side}\ (\pm 1)} \times f^* \times \text{capital}$$

p ← secondary model

- Use the calibrated secondary probability as is
- q = 1 – p is the probability the signal is wrong

β ← triple barrier

- The size of one win divided by the size of one loss
- Already determined when the barrier widths are set

s ← primary model

- The Kelly fraction sets only the size
- The buy or sell side comes from the primary model's sign

The odds β come from the widths of the triple barrier

The multiples of volatility σ at which the lines are drawn are the ratio of win size to loss size

$$\beta = \frac{\text{profit width}}{\text{stop width}} = \frac{k_u\,\sigma}{k_d\,\sigma} = \frac{k_u}{k_d}$$

Example

- Profit-taking at 2σ and stop-loss at 1σ give odds of 2
- Symmetric barriers (1σ, 1σ) give odds of 1 and a Kelly fraction of 2p – 1
- Only once the odds are set can the secondary probability become a fraction

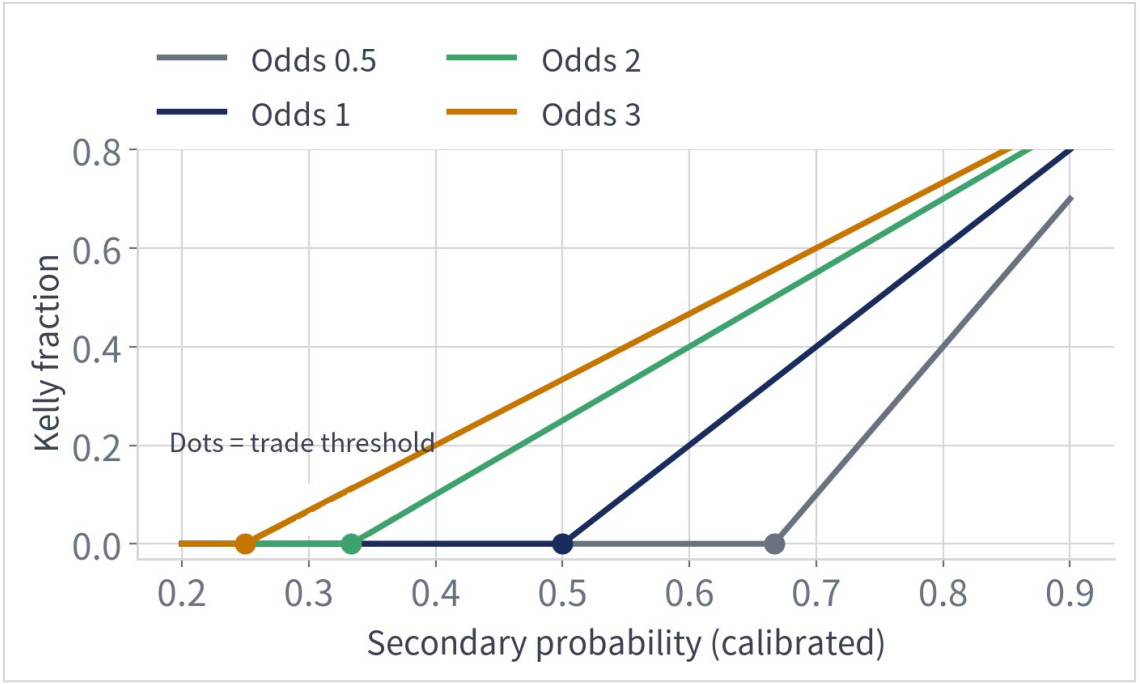


The higher the odds, the lower the probability at which trading is worthwhile

The condition for a positive Kelly fraction yields the trade threshold — whether to trade and how much are one rule

$$f^* > 0 \iff \beta p - (1-p) > 0 \iff p > \frac{1}{1+\beta}$$

Odds β	Trade threshold 1/(1+β)	Fraction at probability 0.6
0.5	0.67	0 (no trade)
1	0.50	20%
2	0.33	40%
3	0.25	47%



The secondary model filters signals below the threshold; above it, higher probability means a larger bet

In the earlier example, a signal with probability 0.6 bets 20% of capital

Kelly fraction by secondary probability under symmetric (odds 1) and asymmetric (odds 2) barriers

Secondary probability	Odds 1: Kelly	Odds 1: half Kelly	Odds 2: Kelly	Odds 2: half Kelly
0.40	0 (no trade)	0	10.0%	5.0%
0.45	0 (no trade)	0	17.5%	8.8%
0.55	10.0%	5.0%	32.5%	16.3%
0.60	20.0%	10.0%	40.0%	20.0%
0.70	40.0%	20.0%	55.0%	27.5%

How to read

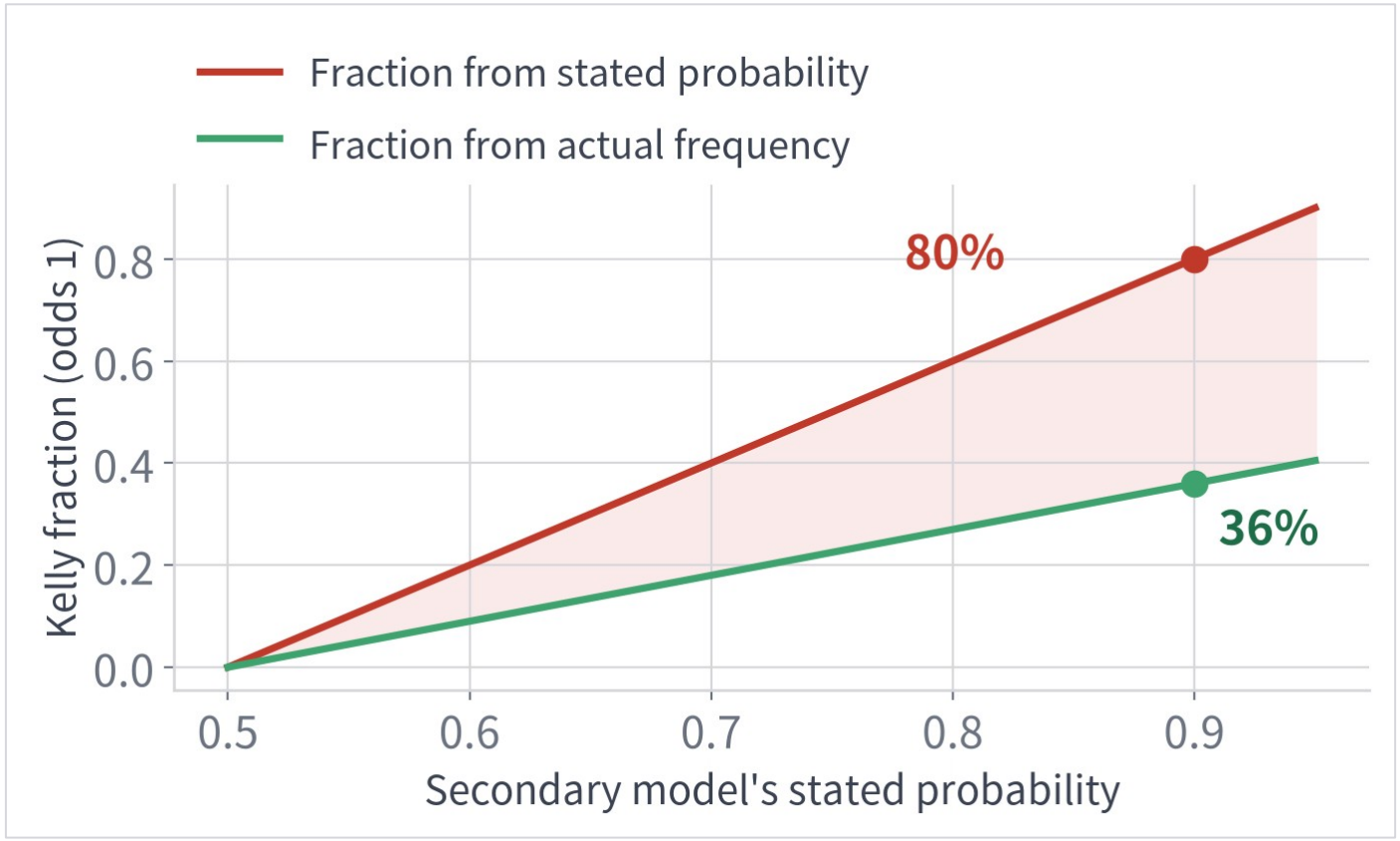
- The same probability 0.45 is filtered at odds 1 but bets 17.5% at odds 2
- In practice, use the half-Kelly columns and set a separate per-asset fraction cap

Feed in uncalibrated probabilities and Kelly overbets by as much as the model is overconfident

The earlier calibration curve — a model that says 0.9 while the actual frequency is 0.68 (※ illustrative data)

What happens

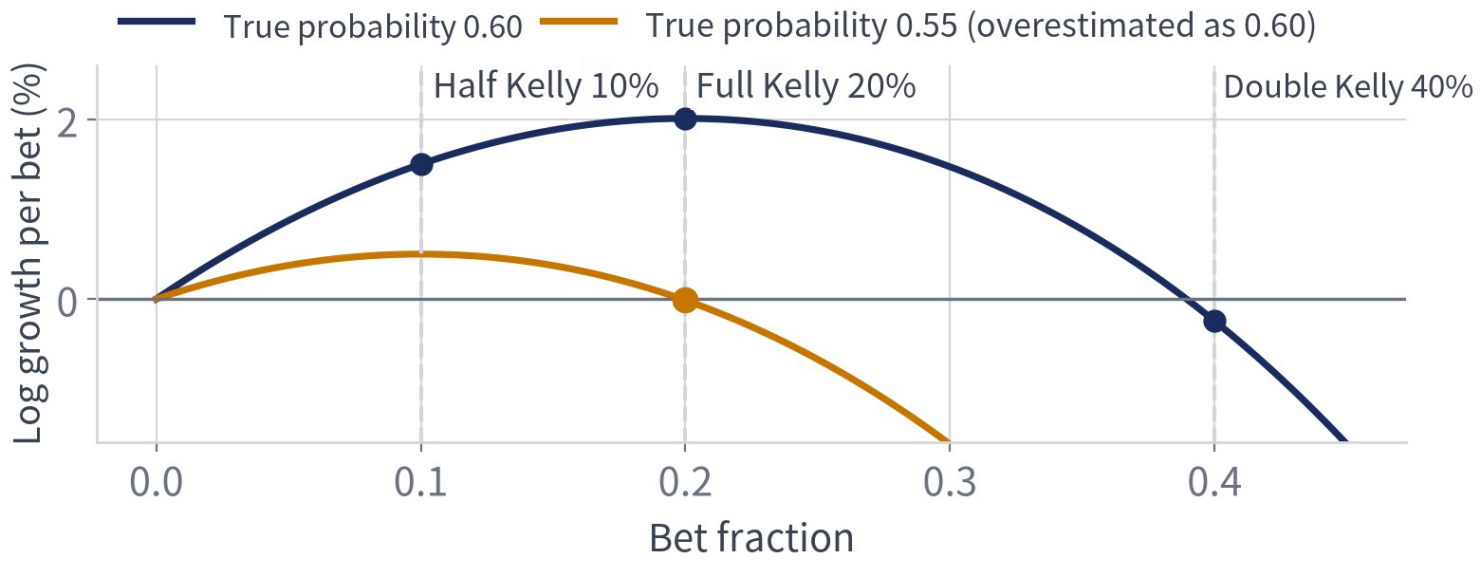
- The stated 0.9 gives a fraction of 80%; the actual frequency 0.68 gives 36%
- Every bet is too large by the red area
- So always calibrate the secondary probability before it goes into Kelly



Even a small overestimate of probability takes full Kelly's growth rate to zero

The Kelly fraction is the f that maximizes long-run log growth g(f) (Thorp) — probability 0.6, odds 1

$$\underbrace{g(f)}_{\text{log growth per bet}} = p\,\ln(1+\beta f) + q\,\ln(1-f)$$

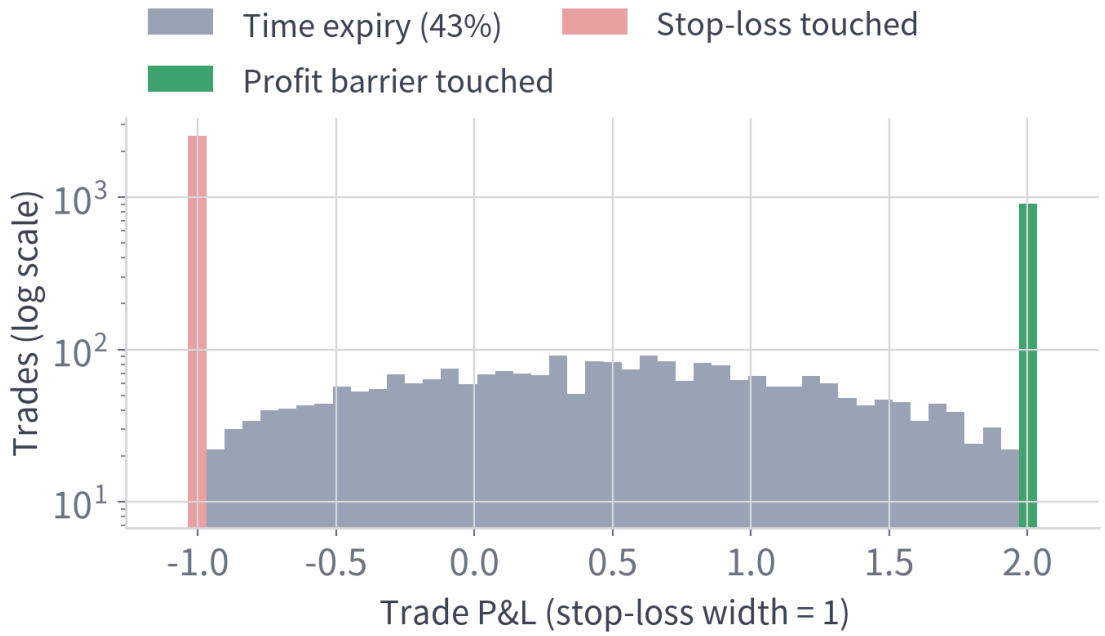


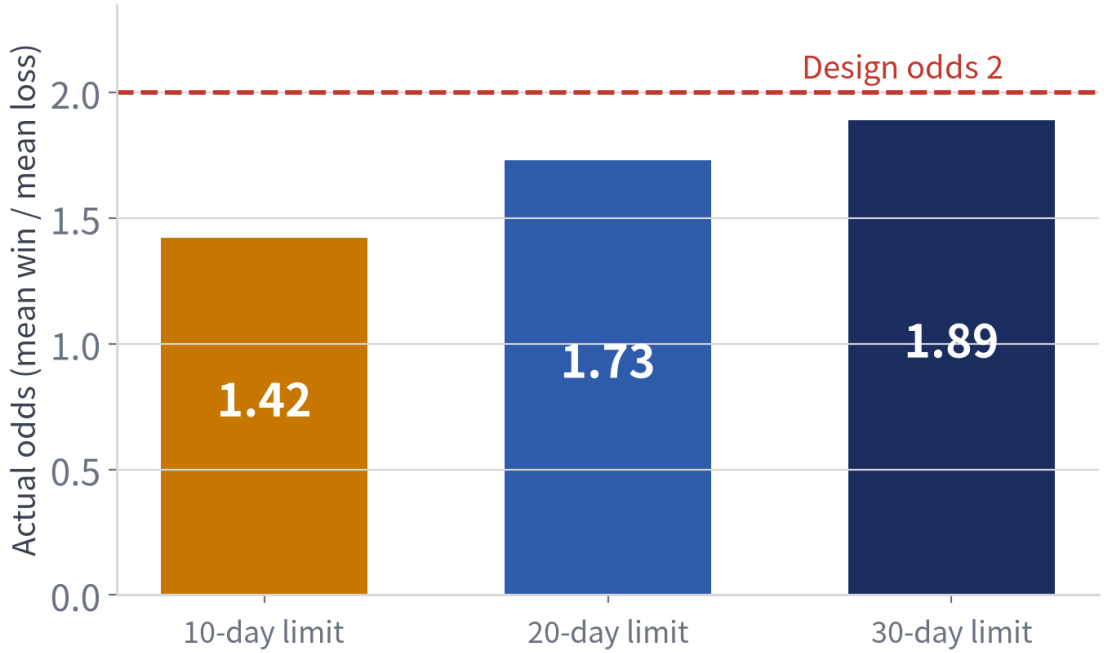
In numbers

- Full Kelly 20%: 2.01% per bet
- Half Kelly 10%: 1.50%, 75% of the growth at half the risk
- Double Kelly 40%: −0.24%, wealth shrinks
- True probability 0.55 believed to be 0.60, betting 20%: growth 0

The more trades expire on time, the further actual odds fall below the design

Profit-taking 2σ, stop-loss 1σ (design odds 2), simulation of 6,000 random price paths (※ illustrative data, daily std = 0.5 × stop-loss width)





With a 10-day limit, 43% of trades touch no line and end with a small P&L, so actual odds are only 1.42

Re-measure odds from actual P&L in the training period, not from the design value

With the wrong odds, the fraction is wrong even if the secondary probability is right

$$\hat{\beta} = \frac{\text{mean win in training period}}{\text{mean loss in training period}}$$

Signal with probability 0.45	Odds used	Kelly fraction
Using the design value	2.00	17.5%
Using the actual value (10-day limit)	1.42	6.3%

Practical remedy

- The design value overbets the same signal by nearly three times
- A longer limit brings actual odds closer to the design value but ties up capital longer

AFML's bet-sizing formula is an S curve close to the Kelly line

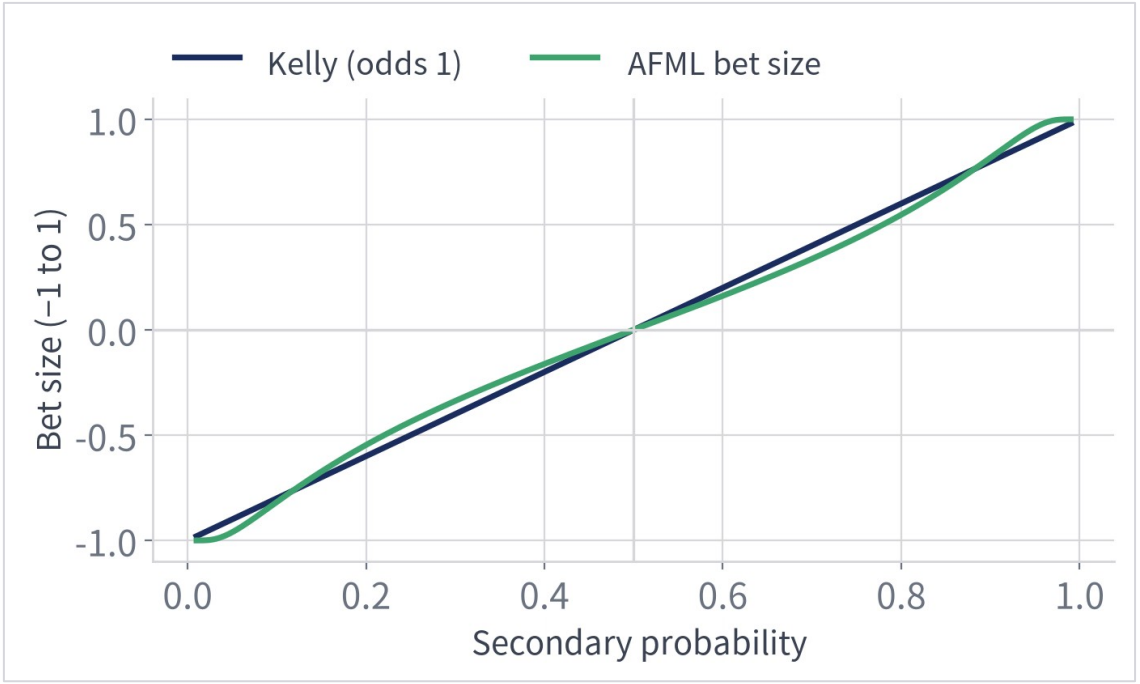
López de Prado (2018), Ch. 10 — turns the secondary probability into a test statistic z, then applies the normal CDF Φ

$$z = \frac{p - \frac{1}{2}}{\sqrt{p(1-p)}} \qquad \underbrace{m}_{\text{bet size}} = 2\,\Phi(z) - 1$$

Probability	Kelly 2p – 1	AFML m
0.55	10.0%	8.0%
0.60	20.0%	16.2%
0.70	40.0%	33.7%

Difference

- Slightly more conservative than Kelly near probability 0.5
- AFML adds averaging of active bets and size discretization on top



The secondary probability becomes an order in five steps

The practical sequence from meta-labeling to a Kelly bet

① Calibrate

Calibrate probabilities on data not used for training

② Estimate odds

Set from barrier widths, then re-measure from actual average win and loss

③ Fraction and threshold

$f^* = (\beta p - q)/\beta$; at probability $1/(1+\beta)$ or below, do not trade

④ Shrink and cap

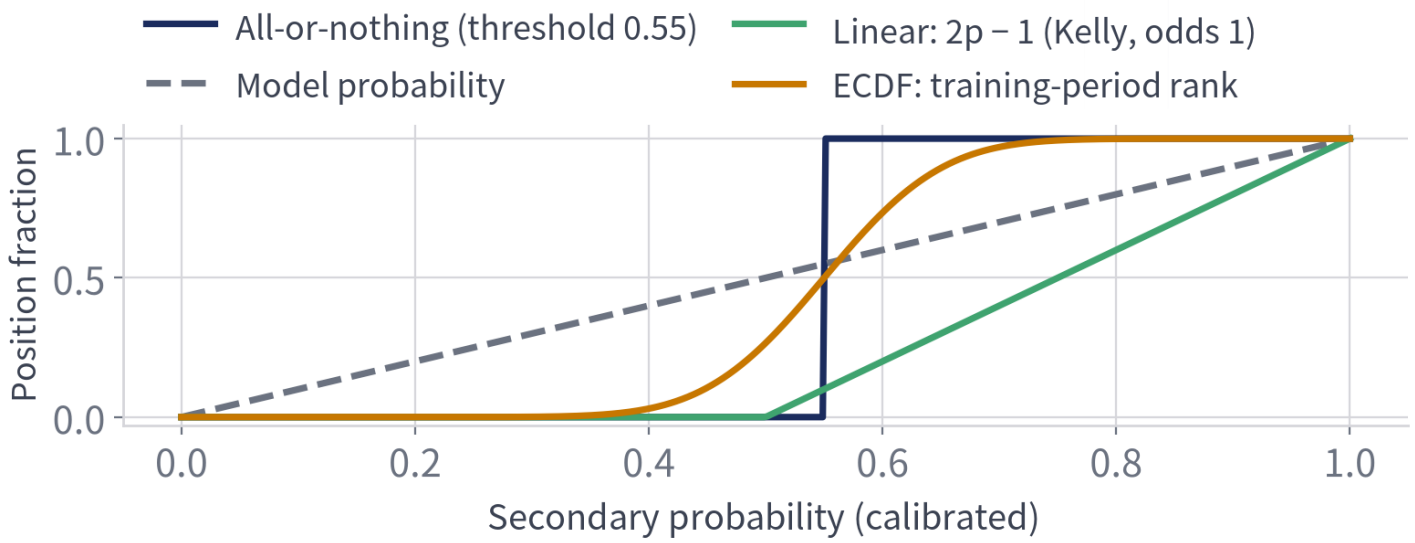
Cut to half Kelly or less and set a per-asset cap

⑤ Apply side

Multiply by the primary model's sign to place a buy or sell order

There are several ways to turn a probability into a fraction

Position-sizing methods compared by Meyer, Barziy and Joubert (2023) (※ ECDF curve is illustrative)



Methods

- All-or-nothing: full size at or above the threshold
- Model probability: p as the fraction
- Linear: normalized by training-period min and max (2p – 1 if the range is 0.5–1)
- ECDF: rank within the training period
- SOPS: a sigmoid that maximizes the Sharpe ratio
- Calibration matters especially for fixed rules

04

PART 4 · Implementation

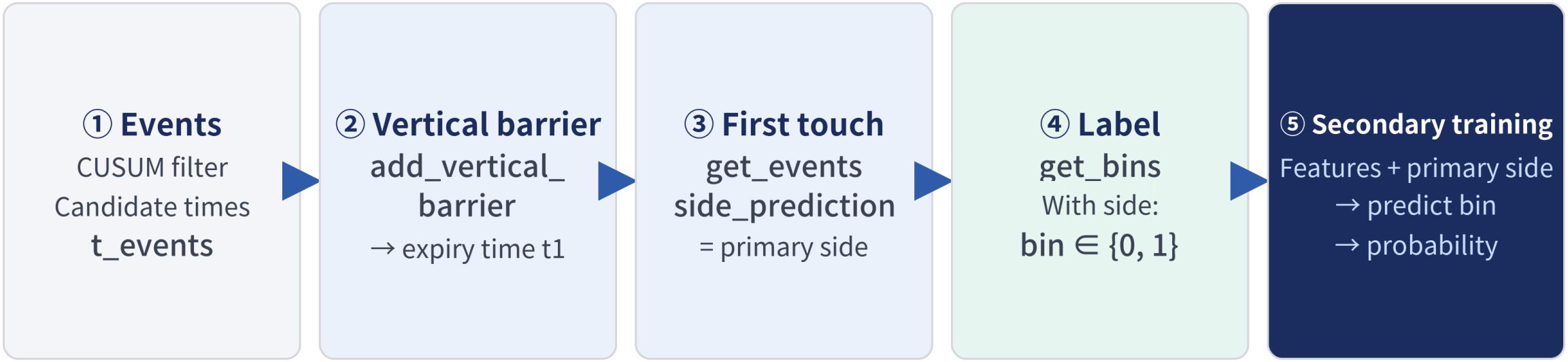
Implementing Meta-Labels

Have Claude Code build the triple barrier and meta-labels with three mlfinlab functions

Meta-labels are built in five steps

Select events, attach limits, find the first touch, assign labels, train the secondary model

Inputs: close, daily volatility (target), profit-taking/stop-loss widths (pt_sl), primary side (side)



Function names follow mlfinlab — they implement Snippets 3.4, 3.6 and 3.7 of Advances in Financial Machine Learning

Have add_vertical_barrier create the holding limit for each event

From each event, finds the first price bar after the set period — Snippet 3.4; the result is the t1 argument of get_events (optional)

[Input] Prompt · Claude Code (Code tab)

```
In the meta-label-lab folder, do the following.
1. Use yfinance to download SPY daily closes from
   2015-01-01 to 2024-12-31 and save them as close.
2. Compute daily volatility (exponentially weighted
   std, span=100). Run a symmetric CUSUM filter with
   its mean as the threshold to get t_events.
3. Create the vertical barrier t1 with
   add_vertical_barrier(t_events, close, num_days=10).
4. If mlfinlab is unavailable, implement AFML
   Snippet 3.4 as a function with the same name.
5. Show the first 5 rows and the counts of t_events
   and t1.
```

What to check

- Is every t1 later than its event time?
- Is the gap at least 10 calendar days? Weekends roll to the next trading day
- Did the count drop because events at the end of the sample have no t1?

Pass the primary side to get_events to build events for meta-labeling

pt_sl: profit-taking and stop-loss widths (0 disables), target: base for the widths, min_ret: minimum target, side_prediction: primary side — Snippet 3.6

[Input] Prompt · Claude Code (Code tab)

```
Next, build events for meta-labeling with get_events.
1. Use a 20-day/50-day moving-average crossover as the
   primary model. Pass its side at each t_events time
   (buy +1, sell −1) as side_prediction.
2. Use the daily volatility above as target.
3. pt_sl=[2, 1]: profit-taking 2σ, stop-loss 1σ.
4. min_ret=0.005: skip targets smaller than this.
5. Use the earlier t1 as vertical_barrier_times.
6. If mlfinlab is unavailable, implement AFML
   Snippet 3.6.
7. Save the result as events.csv and show the first
   5 rows and the count.
```

What to check

- Does events have the t1, trgt and side columns?
- Is the pt_sl order right? Element 0 is profit-taking, element 1 stop-loss
- Does side use only prices up to the event time?
- How many events did min_ret remove?

Have get_bins produce 0/1 meta-labels

No side: bin ∈ {−1, 1}, labeled by price action; with side: bin ∈ {0, 1}, labeled by P&L (meta-labeling) — Snippet 3.7

[Input] Prompt · Claude Code (Code tab)

```
Next, build the meta-labels and train the secondary model.
1. Create bin ∈ {0, 1} with get_bins(events, close).
   If mlfinlab is unavailable, implement AFML
   Snippet 3.7.
2. Features: daily volatility, 20-day return, primary side.
3. Split by time: first 70% train, last 30% validation.
   Drop training samples whose holding period crosses
   the boundary.
4. Train a random forest (500 trees) as the secondary model.
5. Use predict_proba for bet size, primary side for sign.
6. Compare precision and recall with the primary model
   alone in a table.
```

What to check

- Does bin contain only 0 and 1?
- Do no holding periods overlap across training and validation?
- Did precision rise and recall fall?
- Does the sign come from the primary side, not the secondary model?

05

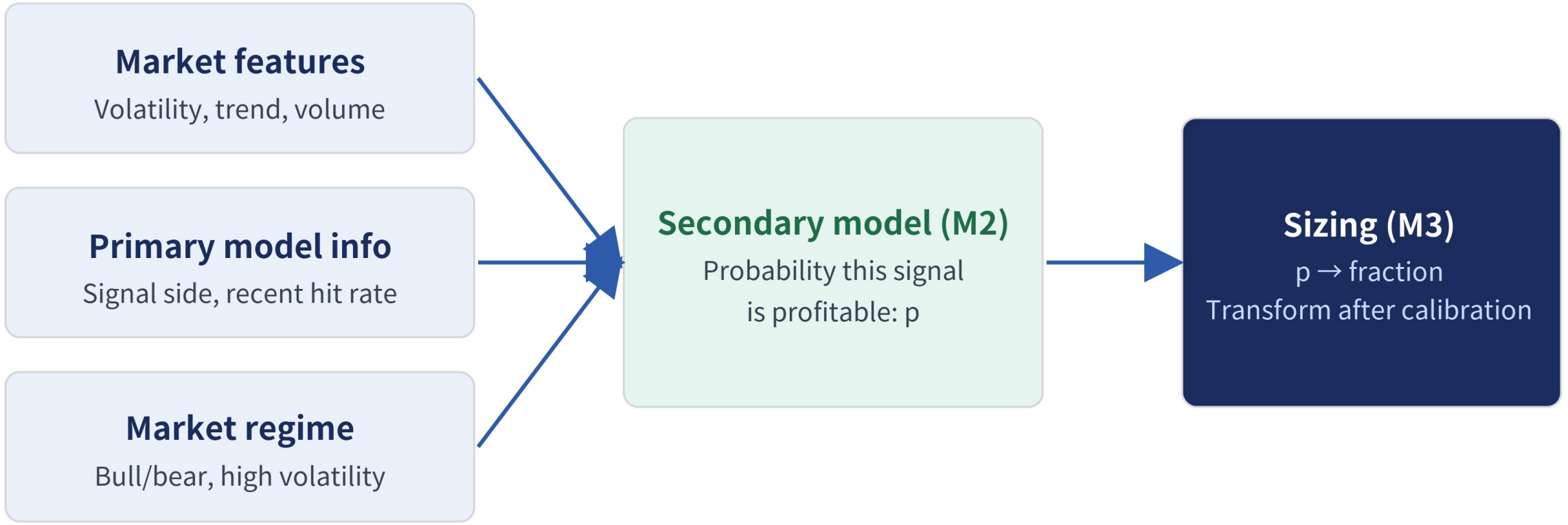
PART 5 · Recent Research

Recent Research and Cases

Frameworks and experimental results from meta-labeling research since 2022

The secondary model looks at market state and the primary model's record together

Joubert's (2022) three-stage framework — M1 sets the side, M2 the profit probability, M3 the size



Some architectures split the secondary model into several

Meyer, Joubert and Alfeus (2022), “Meta-Labeling Architecture”

Long/short split

Separate secondary models for buy and sell signals, each with its own features

Sequential (SMLA)

Secondary models form a chain; each takes the previous model's output

Conditional (CMLA)

A dedicated secondary model per market regime, with results combined

Inverse meta-labeling

Revise the primary model using the secondary model's feature importance, then reapply

When to split

- Split the secondary model when longs and shorts behave differently, or when primary signal quality shifts sharply with the market regime

The more regimes and nonlinearity, the greater the edge of ensemble secondary models

Thumm, Barucca and Joubert (2023), “Ensemble Meta-Labeling”

Bagging type

- Simple models (linear discriminant, perceptron, decision tree) are trained on bootstrap samples and combined by voting or weighting
- Reduces variance and overfitting

Boosting type

- Models of one kind are trained in sequence, each correcting the previous model's errors
- Reduces bias and captures nonlinear interactions

Conclusion

- The gain from ensembles grows as data span more market regimes and patterns are more nonlinear. The combination method (voting, weighting, stacking) is also a model-selection criterion

The meta-label filter cut drawdown but could not rescue a losing strategy

RSI primary model + random forest secondary model, EURUSD 1-hour bars, out of sample 2022–2024

Method	Trades	Win rate	Payoff ratio	Total P&L (pips)	Max drawdown (pips)
RSI only	576	49.3%	0.93	−625	−1,392
Meta-label filter	121	43.0%	0.72	−577	−593
Filter + probability sizing	121	43.0%	0.72	−86	−96

What happened

- Filtering signals below probability 0.55 removed 79% of trades
- The secondary model's out-of-sample accuracy was 48%, worse than a coin flip

Lesson

- The drawdown reduction came from lower exposure, not predictive power
- If the primary model has no edge, the secondary model cannot save it

Meta-labeling results depend on the primary model and the validation method

Four things often missed in practice

It creates no new signals

- The secondary model only filters primary signals. It cannot recover opportunities the primary model missed

Labels overlap

- Labels with overlapping holding periods are not independent. Remove the overlaps in cross-validation (purged k-fold)

Calibrate probabilities

- Uncalibrated probabilities in the Kelly formula overbet by the degree of overconfidence

Evaluate with costs

- Filtering trades also cuts costs. Compare after-cost performance, not precision

The primary model sets the side; the secondary model sets the trade and the size

Three things to remember from meta-labeling

1

Build labels with the triple barrier

The outcome is whichever is touched first: volatility-scaled profit-taking and stop-loss barriers or the holding limit

2

The primary model casts a wide net; the secondary model picks the real signals

In the example, precision rose from 40% to 60% with recall held at 75%. Losing trades fell from 60 to 20

3

Size with calibrated probabilities

At odds 1 the Kelly fraction is 2p − 1. The whole system works only if the primary model has an edge

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- Joubert, J. F. (2022). Meta-Labeling: Theory and Framework. Journal of Financial Data Science, 4(3), 31–44.
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